PRALIDOXIME

Organophosphate poisoning

Give with atropine

CHEMICAL NAME 1-methyl-2-[(E)-(hydroxyimino)methyl]pyridinium (chloride or mesilate) — 2-PAM

CLASS Cholinesterase reactivator (oxime) — antidote

TRADE NAMES Protopam · 2-PAM · Pralidoxime chloride injection · ATNAA (with atropine)

THERAPEUTIC USES / INDICATIONS

- Organophosphate insecticide poisoning (adjunct to atropine) — especially with nicotinic features (muscle weakness, fasciculations).
- Nerve agent poisoning (sarin, VX).
- Not effective for carbamate poisoning (debated).

ADMINISTRATION

- **IV** loading dose over 20–30 min, then continuous infusion; or **IM** (autoinjector in field).
- Always with atropine (atropine treats muscarinic, pralidoxime nicotinic effects).
- Give early before enzyme "ageing".

DOSAGES

- **Adult:** 1–2 g IV over 15–30 min, then 8–10 mg/kg/h infusion (max 650 mg/h) until recovery (often days).
- **Alternative (US label):** 1–2 g IV, repeat after 1 h if weakness persists, then every 10–12 h as needed.
- **Children:** 20–50 mg/kg loading (Québec: max 1 g), then 10–20 mg/kg/h.

PHARMACOKINETICS

- **Onset:** minutes (IV).
- Does not cross blood–brain barrier (quaternary).
- Excreted in urine. **t½:** 1–3 h (longer in renal impairment).

PHARMACODYNAMICS

- Binds organophosphate attached to **acetylcholinesterase** and removes it, reactivating the enzyme; effective before "ageing" (permanent bond) occurs.
- Mainly improves nicotinic effects (muscle weakness, respiratory muscle paralysis).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Blurred vision, diplopia, dizziness.
- Headache, nausea.
- Tachycardia, hypertension.

ADVERSE EFFECTS (SERIOUS)

- Transient neuromuscular blockade (rapid injection) → muscle weakness, laryngospasm.
- Cardiac arrest with rapid IV.
- Worsening myasthenia gravis.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** myasthenia gravis (on anticholinesterases), renal impairment.

KEY INTERACTIONS

- **Atropine:** synergistic (used together).
- **Neostigmine, pyridostigmine:** antagonism; can precipitate myasthenic crisis.

NURSING MANAGEMENT

- Decontaminate patient (remove clothes, wash skin) wearing PPE.
- Airway and breathing support; titrate atropine to drying of secretions.
- Monitor muscle strength, neck flexion, respiratory effort (intermediate syndrome 24–96 h), SpO_2, ECG.
- Infuse slowly; monitor BP and HR.

PATIENT EDUCATION

- Store pesticides safely; use protective equipment when spraying.
- Seek help urgently for exposure symptoms (sweating, salivation, vomiting, weakness).

KEY POINTS

- Oxime; reactivates acetylcholinesterase.
- 30 mg/kg load then 8 mg/kg/h.
- Organophosphate poisoning with atropine.
- Give early, slowly (before ageing).

SOURCES CHECKED **[996]** Québec antidote guide **[997]** Protopam Chloride, IM, IV, SC (US label)

Full references p. 432–457